MAEG4060 Virtual Reality Systems and Applications

Course Project – Stage 1 Proposal (10%)

Submission deadline: 23 Jan 2026

Project Title: Interactive Physical Simulation of Off-Road Vehicle Dynamics in a Mountainous Terrain

Group Members:

- **Guo Tsun Hei (1155186272):** 3D Modeling, Texturing, Rigging, and Animation
- **Yim Fu Chong (1155176974):** Programming, Physics Engine, and System Integration

---

1. Project Theme and Objectives

The objective of this project is to design and develop a high-fidelity, interactive 3D "Physics Sandbox" that simulates the mechanical relationship between a rigid-body vehicle and a complex, non-planar terrain.

In strict adherence to the course requirements, this application is **not a video game**; it contains no missions or scoring systems. Instead, it serves as a technical demonstration of:

1. **Kinematic Terrain-Following:** Real-time calculation of vehicle orientation (Pitch/Roll) based on heightmap gradients.
2. **Momentum & Friction Simulation:** Accurate modelling of acceleration, braking, and rolling resistance.
3. **Dynamic Environmental Interaction:** A living environment featuring hierarchical animations and a time-of-day system.

The application will be developed in **Visual C++** using the **DirectX 9.0c** Fixed Function Pipeline.

---

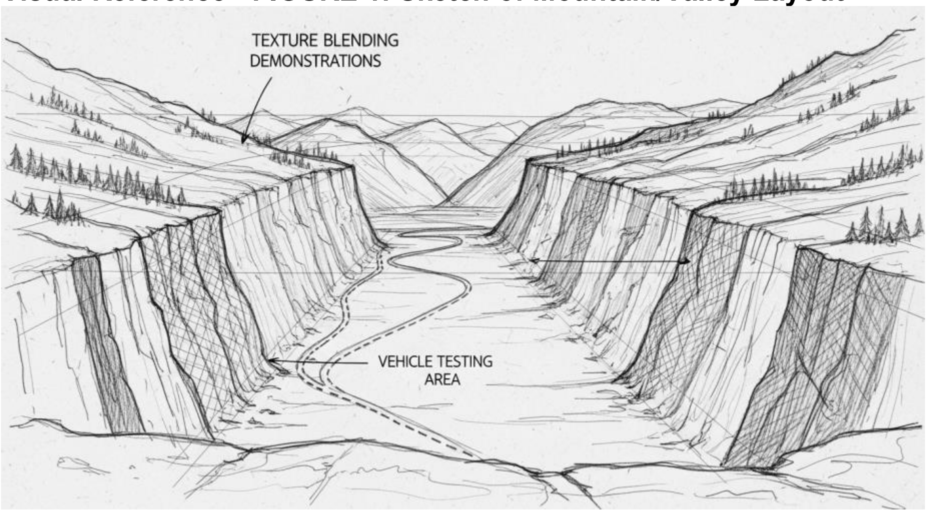
2. Detailed Model Specifications (20%)

Submission deadline: 27 Feb 2026

All assets will be exported in the **.X file format** to ensure compatibility with DirectX frame hierarchies. The project utilizes a **Left-Handed (LH) Coordinate System** (consistent with standard D3DXMatrixLookAtLH functions).

2.1 Landscape/Terrain

- **Geometry:** Generated via a 1024x1024 Heightmap, providing a high-resolution mesh for smooth physics calculations.
- **Surface Shading:** Implementation of **Texture Splatting**. A custom algorithm will blend "Lush Grass" textures on flat plains with "Cragged Rock" textures on slopes exceeding a 30-degree incline.
- **Visual Reference - FIGURE 1: Sketch of Mountain/Valley Layout**

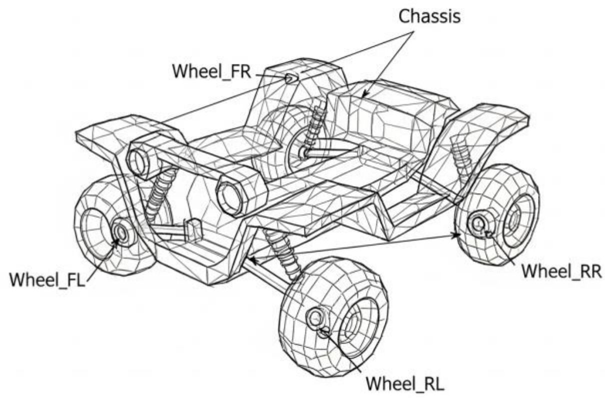


Caption: Sketch illustrating the elevation variance, including the central valley for vehicle testing and the steep cliffs for texture blending demonstrations.

Source: https://sketchfab.com/3d-models/half-dome-terrain-landscape-86ef075dc7d9414583ac2180e1e651a3

2.2 Controllable Vehicle (Jeep/ATV)

- **Hierarchy:** The model features a parent body mesh with four independent wheel nodes. This allows for procedural rotation and steering transformations relative to the chassis.
- **Efficiency:** Poly-count is optimized at approximately 3,000 triangles to ensure high-frequency collision detection without CPU bottlenecking.
- **Visual Reference - FIGURE 2: Sketch of Vehicle Model**

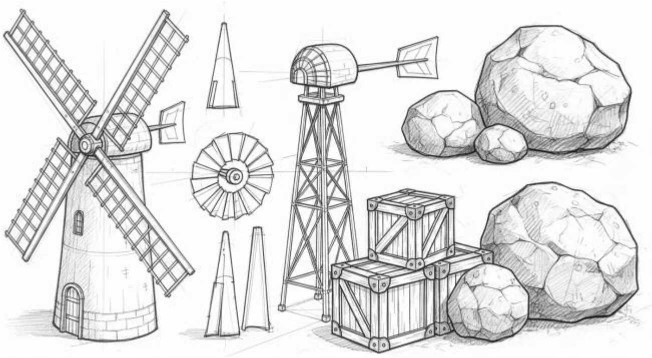


Caption: Technical sketch showing the hierarchical structure (Wheel_FL, Wheel_FR, etc.) and the low-poly chassis design.

Source:https://sketchfab.com/3d-models/atv-from-game-pure-f2e454611ebd4151b9d7aa31815c35e4

2.3 Animated Environmental Elements (20%)

- **Maya Keyframed Animation:** To satisfy the group project requirement, a Windmill prop will be modeled and animated **within Maya**. The sail rotation will be exported as an AnimationSet in the .X file and played back in the engine using ID3DXAnimationController.
- **Procedural Animation:** Vehicle wheels will rotate dynamically based on the calculated linear velocity ($v = \omega r$) to ensure visual fidelity during acceleration and braking.
- **Visual Reference - FIGURE 3:** Sketch of Windmill/Props
  - *Caption: Sketch of the windmill sub-mesh components (target for Maya animation) and environmental props like crates and boulders.*



Caption: Sketch of the windmill sub-mesh components and environmental props like crates and boulders.

Source: https://sketchfab.com/3d-models/rock-boulder-game-ready-asset-0079cb0e717c4aeebe6bca21ec21b0d9

Source: https://sketchfab.com/3d-models/windmill-fe13862b9e3c4dc887d10e297d86f17f

Source: https://sketchfab.com/3d-models/wooden-crate-3c0259d17f3b4306890cdc809b545670

3. Implementation of Effects and Physics

3.1 Movement & Control (5%)

- **Input Handling:** Integration of **DirectInput** for responsive WASD keyboard control and **XInput** for analog gamepad support (trigger-based throttle).
- **Dynamics:** Implementation of inertia and rolling friction constants to simulate realistic deceleration when the throttle is released.

3.2 Terrain Navigation (5%)

- **Bilinear Interpolation:** To prevent "snapping," the vehicle's Y-height will be calculated by interpolating between the four nearest heightmap grid points.
- **Surface Normal Alignment:** The vehicle's "Up" vector will be synchronized with the terrain's surface normal (calculated via the cross-product of the current triangle vertices).

3.3 Collision Detection & Response (10% + 10% Bonus)

- **Static Collision: AABB (Axis-Aligned Bounding Boxes)** will be used for rectangular props (crates), while **Bounding Spheres** will be used for irregular objects (boulders).
- **Bonus Objective: Dynamic-to-Dynamic Collision.** We will implement a momentum transfer system where the vehicle can displace "loose" boulders, causing them to roll down inclines and interact with other world objects.

3.4 Physics-based Simulation & Natural Phenomena (10%)

- **Gravity:** A constant downward force vector **($g = 9.81$ units/s^2)** applied to all non-static entities.
- **Day/Night Cycle:**
  - **Solar Rotation:** The D3DLIGHT9 directional light source will orbit the scene using **sin(time)** and **cos(time)** functions to update its direction vector every frame.
  - **Atmospheric Scattering:** The background clear color and light diffuse/ambient properties will transition from bright amber (Day) to deep indigo (Night).

3.5 Visual Enhancements & User Guide (5%)

- **On-Screen Display (OSD):** Utilizing ID3DXFont to provide a real-time "Debug HUD" showing current velocity, world coordinates (X, Y, Z), and the current state of the Day/Night cycle.

---

4. Work Division

Task Category	Primary Member	Technical Detail
Stage 2: Assets	Guo Tsun Hei	Maya modeling, UV Unwrapping, .X export, hierarchical rigging, and creating keyframe animations in Maya.
Stage 3: Coding	Yim Fu Chong	DX9 Device initialization, Heightmap parser, Physics engine, Collision logic, and Day/Night system.
Integration	Both	Debugging mesh loading errors and tuning physics constants for "feel."

---

5. Final Report Planning (15%)

Submission deadline: 10 April 2026

The final submission will include:

1. **The Executable & Source:** A completed, compiled Visual C++ application.
2. **User Guide:** Detailed instructions for controls and hardware requirements.
3. **Technical Description:** A brief description of the mathematical techniques used for the physics engine, collision detection, and terrain navigation.

References & Asset Credits

1. Half Dome Terrain Landscape

- **Author: Studio Lab**
- **License:** Creative Commons Attribution (CC BY 4.0)
- **Source:** Sketchfab
- **URL:** https://sketchfab.com/3d-models/half-dome-terrain-landscape-86ef075dc7d9414583ac2180e1e651a3
- **Usage:** Reference for high-resolution terrain geometry and heightmap generation.

2. Controllable Vehicle (ATV)

- **Title:** ATV from game "PURE"
- **Author: barking_dogo**
- **License:** Creative Commons Attribution (CC BY 4.0)
- **Source:** Sketchfab
- **URL:** https://sketchfab.com/3d-models/atv-from-game-pure-f2e454611ebd4151b9d7aa31815c35e4
- **Usage:** Main controllable vehicle model (Wheel nodes separated for procedural animation).

3. Rock Boulder (Game Ready Asset)

- **Author: Aparicio Silva 3D**
- **License:** Creative Commons Attribution (CC BY 4.0)
- **Source:** Sketchfab
- **URL:** https://sketchfab.com/3d-models/rock-boulder-game-ready-asset-0079cb0e717c4aeebe6bca21ec21b0d9
- **Usage:** Used for dynamic physics simulation (Bounding Sphere collision).

4. Windmill

- **Author: KIFIR**
- **License:** Creative Commons Attribution (CC BY 4.0)
- **Source:** Sketchfab
- **URL:** https://sketchfab.com/3d-models/windmill-fe13862b9e3c4dc887d10e297d86f17f
- **Usage:** Used for keyframed animation and environmental aesthetics.

5. Wooden Crate

- **Author: Domingos Studios**
- **License:** Creative Commons Attribution (CC BY 4.0)

- **Source:** Sketchfab
- **URL:** https://sketchfab.com/3d-models/wooden-crate-3c0259d17f3b4306890cdc809b545670
- **Usage:** Used for static collision detection (AABB).

Appendix A: Project Execution & Technical Roadmap

1. Core Project Concept

This project is explicitly defined as a **"3D Physics Sandbox"** rather than a traditional video game. It deviates from mission-based or score-based gameplay to focus entirely on technical simulation. The core objective is to demonstrate a realistic mechanical interaction between a rigid-body off-road vehicle and complex, non-planar mountain terrain, governed by fundamental physics laws (gravity, friction, and inertia).

2. Role Responsibilities & Workflow

To ensure efficiency and strict adherence to the **Group Project** requirements, the workload is divided as follows:

Guo Tsun Hei (Assets, Animation & Pipeline)

- **Primary Responsibility:** Creating all visual assets and ensuring they are technically compatible with the DirectX pipeline.
- **Key Deliverables:**
  - **Maya Animation (Critical Requirement):** Creation of the "Windmill" prop with keyframed rotation animations. This serves as the mandatory pre-baked animation asset.
  - **Vehicle Hierarchy:** Modeling the Jeep with four independent wheel nodes (children of the chassis) to allow for procedural rotation control in the engine.
  - **Pipeline Management:** Exporting all scenes (Terrain, Vehicle, Props) into the **.X format** for DirectX 9 integration.

Yim Fu Chong (Engine, Physics & Integration)

- **Primary Responsibility:** Bringing static assets to life through code and physics calculations.
- **Key Deliverables:**
  - **Kinematic Terrain Following:** Implementing algorithms to calculate heightmap gradients, ensuring the vehicle "sticks" to the ground while adjusting Pitch/Roll orientation dynamically.
  - **Physics Simulation:** coding the forces for Gravity ($g = 9.81$), friction, and momentum conservation (preventing instant stops).
  - **Procedural Animation:** Implementing code-driven wheel rotation ($v = \omega r$) that corresponds to vehicle velocity.
  - **Environmental Logic:** Implementing the Day/Night cycle using trigonometric light orbit functions.

3. Technical Highlights & Bonus Objectives

The project aims to demonstrate proficiency in three specific areas:

- **Advanced Texturing:** Implementation of **Texture Splatting**, where the terrain shader automatically switches between "Grass" and "Rock" textures based on a slope threshold (>30 degrees).
- **Hybrid Collision Strategy:** Demonstrating knowledge of multiple algorithms by using **AABB** (Axis-Aligned Bounding Boxes) for crates and **Bounding Spheres** for boulders.
- **Bonus Objective (Dynamic-to-Dynamic Interaction):** A momentum transfer system where the vehicle can collide with "loose" boulders, causing them to physically roll down the terrain, rather than acting as static walls.

4. Project Status (as of Jan 20, 2026)

The Stage 1 Proposal is complete. The team has established the technical scope and asset requirements. The workflow is finalized: **Maya** will be used for high-fidelity modeling and keyframed animation, while **Visual C++ / DirectX 9** will handle the real-time physics simulation and procedural effects. The team is prepared for full development immediately following this submission.